

Executive Summary

PASS RATE 25% target 90%	TOTAL TESTS 168 executed	PASSED 43 26% of total	FAILED 125 74% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 25% | Target: 90% | Gap: +65%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 25% pass rate across 168 tests, substantially below the 90% target threshold. The API Automation suite shows 21 of 80 endpoints functioning correctly while Backend Smoke Validation indicates 22 of 88 core functionalities are operational. This suggests widespread backend service reliability issues affecting both standard API operations and critical system pathways. The consistent failure pattern across both test suites indicates systemic backend challenges rather than isolated endpoint issues, requiring immediate technical intervention to restore service stability and meet operational requirements.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	74	21	53	28%	FAIL
2	BE Smoke Validation	88	22	66	25%	FAIL
TOTAL		168	43	125	25%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#46	04 Jul	168	25%	125	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	43	25.6%	180ms	Within SLA
Requests Failed	125	74.4%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	6%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the most frequently failing endpoints to identify common failure patterns	Immediate	P1
2	Implement comprehensive backend service health monitoring and logging to capture failure conditions in real-time	3 days	P1
3	Prioritize stabilization of smoke validation endpoints as these represent core system functionality	5 days	P2
4	Review backend service dependencies, database connections, and inter-service communication protocols for potential bottlenecks	1 week	P2
5	Deploy incremental fixes starting with high-impact endpoints while maintaining continuous validation through automated testing	1 week	P2
6	Establish backend performance profiling to identify resource constraints or timeout issues affecting API response reliability	2 weeks	P3

KEY FINDINGS

1	Backend services achieving only 25% reliability across 168 automated validation points
2	API endpoints demonstrating 26% success rate with 59 of 80 endpoints returning incorrect responses or experiencing failures
3	Core smoke validation paths showing 75% failure rate indicating fundamental backend service instability
4	Both API automation and smoke validation suites exhibiting similar failure patterns suggesting systemic backend issues
5	Critical gap of 65 percentage points from the 90% target pass rate requiring urgent remediation

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



28%

